

ABHISHEK CHOTHANI

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EXECUTIVE SUMMARY

Innovative robotics engineer with 2+ years of experience in automation and manufacturing, blending expertise in mechanical design and advanced robotics programming. Proficient in **Python, ROS, Isaac Sim and Drake** with a strong foundation in deep learning, computer vision and deep reinforcement learning for robotics applications. Passionate about leveraging cutting-edge technologies to push the boundaries of robotic capabilities.

EDUCATION

Florida Institute of Technology | Master of Science, Mechanical Engineering | GPA: 3.6 Melbourne, FL | Dec 2024
Specialization in Robotics and Automation, Robotics Manipulation, Deep Learning, Reinforcement Learning
Thesis: End-to-End Learning for a Low-Cost Robotics Arm, advised by [Dr. Ryan White](#) ([Thesis Link](#))
Gujarat Technological University | Bachelor of Engineering, Mechanical Engineering | GPA: 3.3 Gujarat, India | May 2021
Design and Manufacturing, Dynamics and Control, Artificial Intelligence, Mechatronics

EXPERIENCE

Research Assistant | Neural Transmissions ([NETS](#)) Lab Melbourne, Florida | Jan 2024 – Dec 2024

- Developed and evaluated **imitation learning** and **reinforcement learning** for custom 6 DOF robotic manipulators using tools like **PyTorch, Isaac Sim, Isaac Lab** and the **LeRobot framework**.
- Developed a model-based pick-and-place system in **Drake** to compare classical and learning-based robotics.
- Trained and deployed **Action Chunking with Transformers (ACT)**, **Action Diffusion Policies** and **Proximal Policy Optimization (PPO)**, achieving up to 85% success in dynamic manipulation tasks such as stacking, orienting, and sliding.
- Performed comparative analysis of reinforcement learning (e.g., PPO) vs. classical methods, identifying strengths and limitations for industrial-scale deployment. ([Thesis Demonstration Link](#))

Robotics Project Developer Intern | Kennedy Space Center Visitor Complex Cape Canaveral, Florida | June 2024 – Present

- Designed and built an educational replica of the NASA Perseverance Rover for students and visitors, using **Fusion 360, Raspberry Pi**, and **STM32** to demonstrate real-world space robotics concepts.
- Developed a **ROS2**-based control with **C++** featuring real-time motion control for navigation and trajectory tracking.
- Integrated and calibrated IMUs, ultrasonic sensors, and cameras for semi-autonomous operation, enabling obstacle detection, localization, and basic **path planning**.
- Collaborated with educators to design an interactive demonstration of space technologies to the students. ([Project Link](#))

Mechanical Engineer Intern | Jaycon Systems Melbourne, Florida | Sep 2024 – Dec 2024

- Designed and simulated mechanical components of an augmented reality (AR) device using **Fusion 360 and ANSYS**, ensuring reliability and performance under real-world use conditions.
- Built and deployed deep learning-based object detection and tracking systems using **YOLO and OpenCV**, fully integrated into AR hardware and optimized for real-time performance on **NVIDIA Jetson Orin**.
- Enhanced on-device AI inference to support interactive object recognition within AR environments, improving user responsiveness and system autonomy. ([Project Link](#))

Robotics Engineer | BandG Robotics Ahmedabad, India | Jan 2021 – Dec 2022

- Programmed robotic manipulators using **ROS and Python**, enhancing motion accuracy with advanced control algorithms.
- Designed custom robotic end-effectors and mechanical components in **Fusion 360 and SolidWorks**, increasing system adaptability across diverse use cases.
- Refined **CNC manufacturing** workflows by using **Mastercam** to optimize toolpaths and machining parameters, resulting in high-precision and efficient production of robotic parts.

SKILLS

- Programming & Frameworks:** Python, C/C++, PyTorch, ROS (Noetic, 2), RViz, MoveIt, Drake, OpenCV, NumPy, Matplotlib
- Robotics & AI:** Computer Vision, Robotic Manipulation, Path Planning, Navigation, Imitation Learning, Deep Learning, Reinforcement Learning, Sensor Integration (Lidar, Camera, IMU, etc.)
- CAD & Mechanical Tools:** Fusion 360, SolidWorks, ANSYS, FEA (Finite Element Analysis), 3D Printing, Mastercam
- Other Tools:** Git, Linux, Jupyter Notebooks, Isaac Sim, VS Code, Ubuntu, GitHub

PROJECTS

- Bipedal Training of Spot using IsaacLab & PPO (2025)** – ([Project Link](#))
- ROS Noetic Tutorials (2024)** – Step-by-step tutorials for learning ROS Noetic. ([Project Link](#))
- Pan and Tilt Object Tracking System (2024)** – Object Tracking system using YOLO. ([Project Link](#))
- Human Gesture-Controlled Drone (2023)** – Vision-based Drone Control using PX4. ([Project Link](#))
- ARM Cortex-M4 Development (2023)** – Embedded programming with FreeRTOS. ([Project Link](#))